

Formation of Amphiphilic Zwitterionic Thin Poly(SBMA-co-TFEMA) Brushes on Solid Surfaces for Marine Antifouling Applications

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Cite This: *Langmuir* 2024, 40, 3213–3221



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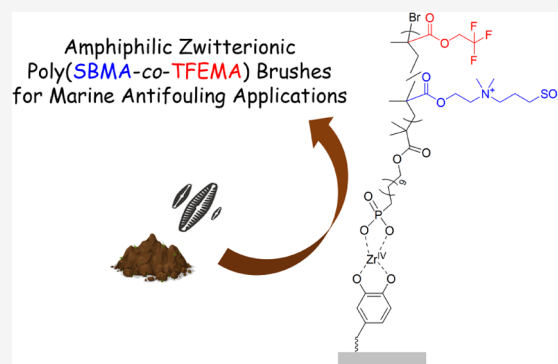


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ABSTRACT: Water molecules can bind to zwitterionic polymers, such as carboxybetaine and sulfobetaine, forming strong hydration layers along the polymer chains. Such hydration layers act as a barrier to impede the attachment of marine fouling organisms; therefore, zwitterionic polymer coatings have been of considerable interest as marine antifouling coatings. However, recent studies have shown that severe adsorption of marine sediments occurs on zwitterionic-polymer-coated surfaces, resulting in the degradation of their marine antifouling performance. Therefore, a novel approach for forming amphiphilic zwitterionic polymers using zwitterionic and hydrophobic monomers is being investigated to simultaneously inhibit both sediment adsorption and marine fouling. In this study, amphiphilic zwitterionic thin polymer brushes composed of sulfobetaine methacrylate (SBMA) and trifluoroethyl methacrylate (TFEMA) were synthesized on Si/SiO₂ surfaces via surface-initiated atom transfer radical polymerization. For this, a facile metal-ion-mediated method was developed for immobilizing polymerization initiators on solid substrates to subsequently form poly(SBMA-co-TFEMA) brushes on the initiator-coated substrate surface. Poly(SBMA-co-TFEMA) brushes with various SBMA/TFEMA ratios were prepared to determine the composition at which both marine diatom adhesion and sediment adsorption can be prevented effectively. The results indicate that poly(SBMA-co-TFEMA) brushes prepared with an SBMA/TFEMA ratio of 3:7 effectively inhibit both sediment adsorption and marine diatom adhesion, thereby exhibiting balanced marine antifouling properties. Thus, the findings of this study provide important insights into the design of amphiphilic marine antifouling materials.



1. INTRODUCTION

Biofouling, which refers to the undesired adsorption of marine organisms to submerged solid surfaces, has a major impact on the maritime industry.^{1–3} The biofouling of a ship hull increases surface friction, leading to a higher fuel consumption,⁴ which, in turn, causes significant greenhouse gas emission. Therefore, various methods are being investigated to prevent biofouling,¹ among which the use of antifouling polymer coatings has received considerable attention owing to its environmental friendliness.^{5–16} Polymers with good hydrophilicity are widely utilized as antifouling coatings because the hydration layer formed on hydrophilic polymer coatings in water serves as a physical barrier to prevent the adherence of marine organisms to the substrate surface.¹⁷ Zwitterionic polymers prepared from monomers such as sulfobetaine (SB), phosphorylcholine, and carboxybetaine (CB) are typical hydrophilic polymers with antifouling properties.^{18–20} These polymers have been successfully applied in surface coating technologies to achieve excellent marine antifouling properties. However, zwitterionic polymer coatings tend to adsorb sediments when exposed to the marine environment.^{21,22} This results in a reduction in the antifouling performance of the coating, necessitating the development of alternative coating materials.

Amphiphilic polymers, which contain both hydrophilic and hydrophobic components, are of considerable interest in various fields owing to their ability to adsorb at interfaces and self-assemble into aggregates.²³ These polymers are widely applied in biotechnology, detergents, agriculture, cosmetics, food, electronics, and pharmaceuticals.^{24–28} Furthermore, by incorporating both hydrophobic and zwitterionic components into polymer coatings, the surface can be rendered “ambivalent” to marine biofoulants, thereby inhibiting biofouling.^{23,29} Amphiphilic zwitterionic polymer coatings are also less prone to sediment adsorption than zwitterionic polymer coatings owing to the decreased fraction of the zwitterionic components, which tend to adsorb sediments.³⁰ Thus, amphiphilic zwitterionic polymers are promising as antifouling coating materials that can overcome the limitations of zwitterionic polymers.²⁹ Hitherto, various amphiphilic

Received: November 29, 2023

Revised: January 20, 2024

Accepted: January 23, 2024

Published: February 5, 2024



zwitterionic polymers have been synthesized, and their antifouling properties have been confirmed.^{29–31} For example, Koschitzki et al. synthesized an amphiphilic zwitterionic polymer containing CB and dicyclopentenyl groups and demonstrated its antifouling properties against both marine diatoms and sediments.²⁹

Polymer coatings are prepared via two approaches: the “grafting-onto” and “grafting-from” approaches.^{32–38} In the “grafting-onto” method, presynthesized polymers are grafted onto the solid surface using specific interactions between the polymer and surface. In contrast, in the “grafting-from” approach, initiators are immobilized on the solid surface, and then, polymer brushes are grown using the initiators. Compared to the “grafting-onto” approach, the “grafting-from” approach offers greater flexibility in the design of the polymer brush and facilitates a higher-chain grafting density. Therefore, various “grafting-from” methods, such as surface-initiated reversible addition–fragmentation chain-transfer polymerization,³⁹ surface-initiated nitroxide-mediated polymerization,⁴⁰ and surface-initiated atom transfer radical polymerization (SI-ATRP), have been developed for producing polymer coatings.⁴¹ Marine fouling depends on the composition (hydrophobic content) of an amphiphilic polymer. Therefore, amphiphilic polymer brushes with various hydrophobic contents must be systematically formed to determine the optimal composition for achieving marine antifouling properties.⁴² SI-ATRP is a suitable technique to achieve this goal because it is compatible with a wide range of monomers and can tolerate high concentrations of impurities.^{41,43} However, there have been a few examples of controlling the composition of amphiphilic zwitterionic polymer brushes specifically via SI-ATRP and then systematically investigating the marine antifouling properties of each composition. Herein, we present a method to control the surface composition of amphiphilic zwitterionic polymer brushes via SI-ATRP. We grew thin polymer brushes composed of sulfobetaine methacrylate (SBMA) and trifluoroethyl methacrylate (TFEMA) on Si/SiO₂ surfaces via SI-ATRP. Specifically, we prepared poly(SBMA-*co*-TFEMA) brushes with various SBMA/TFEMA ratios to determine the optimal ratio that provides effective resistance against both marine diatom adhesion and sediment adsorption.

2. EXPERIMENTAL SECTION

2.1. Materials. Dopamine hydrochloride (DA, 98%), tris(hydroxymethyl)aminomethane hydrochloride (Trizma HCl, 99%), tris(hydroxymethyl)aminomethane (Trizma base, 99%), zirconyl chloride octahydrate (ZrOCl₂·8H₂O, 98%), [2-(methacryloyloxyethyl)dimethyl-(3-sulfopropyl)ammonium hydroxide (SBMA, 95%), 2,2,2-trifluoroethyl methacrylate (TFEMA, 99%), and *N,N,N',N'*-pentamethyldiethylenetriamine (PMDETA, 99%) were procured from Sigma-Aldrich. [11-[(2-Bromo-2-methylpropanoyl)oxy]undecyl]phosphonic acid (initiator, >98%), 2,2'-bipyridyl (>99%), and L-ascorbic acid (>99%) were provided by TCI. Copper(I) bromide (CuBr, 98%), copper(II) bromide (CuBr₂, 99%), and 2,2,2-trifluoroethanol (TFE, >99%) were procured from Alfa Aesar. All chemicals were used as received. The marine benthic diatom, *Amphora coffeaeformis*, was obtained from the Korea Marine Microalgae Culture Center (KMMCC).

2.2. Preparation of Initiator-Coated Solid Substrates. The solid substrates (Si/SiO₂, stainless steel (SS), and glass) were cleaned with ethanol via sonication for 5 min and dried under a stream of nitrogen gas. They were then immersed in a DA solution (10 mM, tris buffer, pH 8.4) for 4 h and washed with deionized (DI) water. The resulting polydopamine (pDA)-coated substrates were immersed in

an ethanolic solution of ZrOCl₂ (5 mM) for 10 min, rinsed with ethanol and DI water, and dried under a stream of nitrogen gas. The substrates were subsequently incubated in an ethanolic solution of the initiator (1 mM) for 10 min, rinsed with ethanol and DI water, and dried under a stream of nitrogen gas.

2.3. Formation of Poly(SBMA-*co*-TFEMA)_y Brushes via SI-ATRP. Before use, the TFEMA monomer was purified using a basic alumina syringe to remove the inhibitors. First, SBMA and TFEMA were dissolved in different ratios in a solvent mixture (TFE/MeOH = 2:1) (Table 1). Next, the monomer solution (1.5 mL) was transferred

Table 1. Final Concentrations of Monomers Used for Preparing Each Type of Polymer Brush

polymer brush	SBMA concentration (mM)	TFEMA concentration (mM)
poly(SBMA)	1750	0
poly(SBMA- <i>co</i> -TFEMA) _y , x/y	875	875
	525	1225
	350	1400
	175	1575

to a 20 mL glass vial containing the initiator-coated solid substrate. The solution was degassed by bubbling nitrogen through it for 30 min. Thereafter, 1.5 mL of a copper/ligand solution consisting of CuBr (10 mg/mL), CuBr₂ (5 mg/mL), and PMDETA (22 μL/mL) was added to the glass vial for polymerization. The copper/ligand solution was also degassed by bubbling nitrogen for 30 min before use. After 6 h of polymerization at 70 °C, the solid substrates were washed with TFE and MeOH and dried under a stream of nitrogen gas.

2.4. Characterization of the Polymer-Coated Substrate. X-ray photoelectron spectroscopy (XPS) was conducted by using a K-Alpha⁺ spectrometer (Thermo Scientific, UK) equipped with an Al Kα X-ray source. The thickness of the organic coating on the solid substrate was measured using an Alpha-SE spectroscopic ellipsometer (J.A. Woollam) in the wavelength range of 390–900 nm at an incidence angle of 70°. The data were fitted to a Cauchy model using the CompleteEASE software (refractive index $n(\lambda) = 1.45 + 0.01/\lambda^2$). Static water contact angle (WCA) measurements (Phoenix-10; Surface Electro Optics, Suwon, South Korea) were performed to determine the hydrophobicity of the coatings. Atomic force microscopy (AFM) images were obtained in the noncontact mode using an XE-7 microscope (Park Systems, Suwon, South Korea). Finally, fluorescence images were obtained using an optical microscope equipped with a fluorescence imaging system (LV100ND; Nikon, Tokyo, Japan).

2.5. Dynamic Marine Diatom Adhesion Test. The diatom suspension was spread on the solid substrates and incubated in a reciprocating shaker (MaXshakeRP2; Daihan Scientific Co.) for 24 h at 50 rpm. The solid substrates were then gently washed with artificial seawater. The fouling behavior of diatoms on the solid substrates was analyzed by using fluorescence microscopy.

2.6. Silt Adsorption Test. Natural sand sediment from the test site at Port Boryeong, South Korea (36°20'15.1"N 126°31'57.5"E), was autoclaved and suspended in sterile filtered seawater at a concentration of 37.5 g/L. The silt adsorption tests were conducted on glass substrates; only one side of the substrates was grafted with polymers. For testing, the substrates were immersed in the sediment suspension and incubated in a reciprocating shaker for 60 min at 50 rpm. They were then gently washed with artificial seawater and dried under ambient conditions. Photometric absorption at 440 nm (UV-1800, Shimadzu, Tokyo, Japan) was measured for each sample surface and compared with that of the nonsilt-adsorbed substrate, as it is known that the glass substrates absorb very little light at a wavelength of 440 nm, while significant light absorption occurs by marine sediment adsorption.^{21,22}

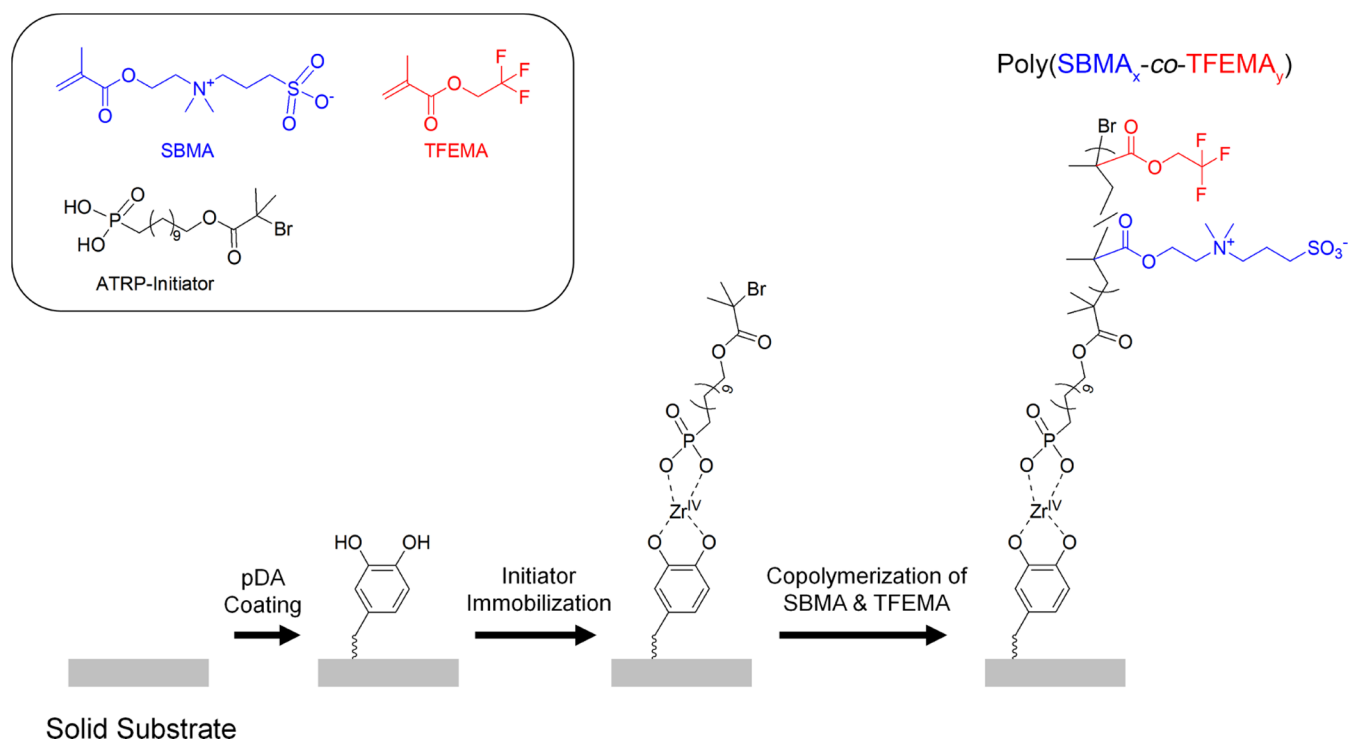


Figure 1. Schematic of the amphiphilic zwitterionic poly(SBMA_x-co-TFEMA_y) brush formation process.

2.7. Statistical Analysis. Three independent samples were prepared. The data are presented as the mean $\pm$ 95% confidence limit. Statistical significance was determined using an unpaired, two-sided Student's *t*-test ($n = 3$). *** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$.

3. RESULTS AND DISCUSSION

To immobilize the ATRP initiator on the solid substrate, we used a metal-mediated coordination approach (Figure 1). The catechol group was used as the site for immobilizing the ATRP initiator.⁴⁴ Among various metal ions, Zr^{IV} was chosen for binding the initiator because of its ability to form stable coordination bonds with various functional groups such as pyrogallol, catechol, carboxylic acid, sulfonate, and phosphonic acid.⁴⁵ First, the solid substrate was pretreated with pDA to introduce catechol groups on the surface.⁴⁴ Next, the pDA catechol groups and phosphonic acid-terminated initiator were cross-linked via Zr^{IV}-mediated multiple coordination bonding to produce initiator-coated solid substrates. Thereafter, amphiphilic polymer brushes were formed on the initiator-coated surface via SI-ATRP. SBMA and TFEMA were chosen as the zwitterionic and hydrophobic monomers, respectively, because the combination of these two monomers has been widely used by researchers in the fields of membrane science and antifouling coatings.^{46–48} A series of amphiphilic zwitterionic thin polymer brushes composed of SBMA and TFEMA were formed on solid substrates by varying the ratio of the two monomers. To investigate the effect of TFEMA on the antifouling performance of the poly(SBMA) brush, polymer brushes with the TFEMA content above that of SBMA were formed. The feed molar ratios of SBMA and TFEMA (x/y) were set as 5:5, 3:7, 2:8, and 1:9.

The initiator-immobilized substrate surface was characterized via WCA measurements and XPS. The WCA of the uncoated Si/SiO₂ surface decreased from 58.8 ± 1.9 to $36.2 \pm 1.6^\circ$ after pDA coating. With the immobilization of the initiator

on the pDA-coated surface, the WCA increased to $78.8^\circ \pm 1.1^\circ$ owing to the long alkyl chain of the phosphonic acid-functionalized initiator (Figure 2a). XPS confirmed the successful immobilization of the initiator (Figure 2b). In addition to the strong Si 2p peaks of the substrate, C 1s and O 1s peaks were observed for the uncoated Si/SiO₂ substrate. After pDA coating, the peak intensity of Si 2p decreased, and a new N 1s peak corresponding to pDA appeared. Initiator immobilization on the pDA-coated surface further changed the surface chemical composition, resulting in the appearance of new Zr 3d, P 2p, and Br 3d peaks. These results indicate that the phosphonic acid-functionalized initiator was successfully introduced on the pDA-coated surface via Zr^{IV}-mediated cross-linking reactions.

The growth of copolymer brushes on the initiator-coated surface was confirmed via spectroscopic ellipsometry. The thickness of the initiator-coated surface (before polymerization) was 9.2 ± 0.3 nm, and the coating thickness increased after SI-ATRP of SBMA and TFEMA on the initiator-coated surface. The copolymer thicknesses obtained via SI-ATRP with four different feed molar ratios of SBMA and TFEMA (SBMA/TFEMA = 5:5, 3:7, 2:8, 1:9) were measured to be 6.0 ± 1.0 , 5.5 ± 1.8 , 4.6 ± 0.7 , and 1.9 ± 0.8 nm, respectively. These results reveal that the copolymer thickness decreased with an increase in the TFEMA content. As a control, poly(SBMA) was grown in the same manner on the initiator-coated surface, and the polymer thickness was measured to be 10.6 ± 1.6 nm, which is higher than the thicknesses of all poly(SBMA_x-co-TFEMA_y) brushes. Despite the decrease in the polymer thickness, an increase in WCA was observed with increasing TFEMA content. The WCAs of the copolymer brushes were 71.1 ± 2.7 , 85.1 ± 4.8 , 100.6 ± 0.4 , and $100.6 \pm 0.5^\circ$ at SBMA/TFEMA ratios of 5:5, 3:7, 2:8, and 1:9, respectively (Figure 3a). Considering that the WCA of poly(SBMA) is less than 10° , amphiphilic copolymer brushes

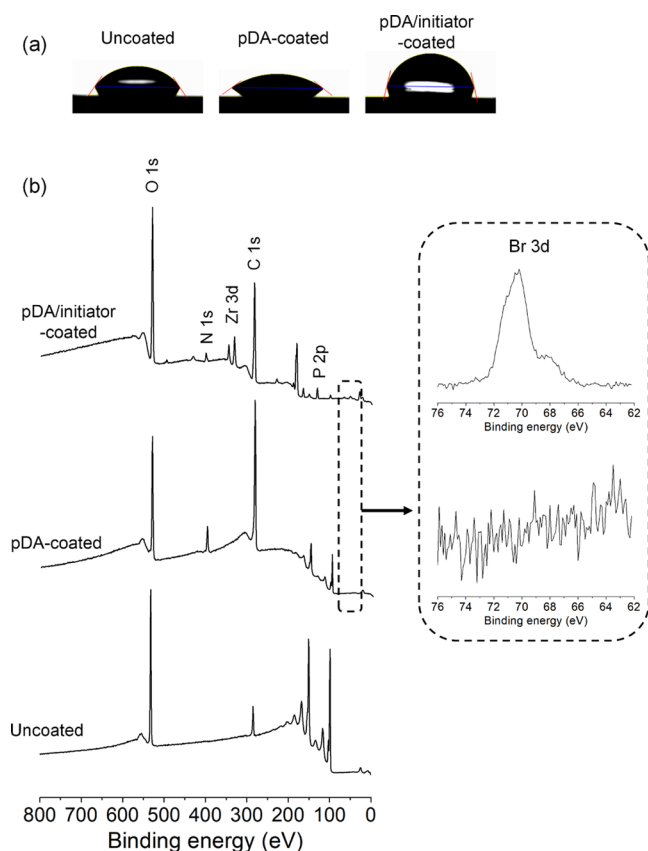


Figure 2. (a) Images showing the water contact angles of the uncoated Si/SiO₂ substrate and Si/SiO₂ substrates coated with pDA and pDA/initiator. (b) Survey and high-resolution (Br 3d) XPS profiles of the uncoated Si/SiO₂ substrate and Si/SiO₂ substrates coated with pDA and pDA/initiator.

with different surface wettabilities were successfully formed via SI-ATRP of SBMA and TFEMA.

The formation of amphiphilic zwitterionic poly(SBMA_{*x*}-co-TFEMA_{*y*}) brushes was also confirmed via XPS (Figure 3b). All poly(SBMA_{*x*}-co-TFEMA_{*y*}) brushes exhibited N 1s (C–N⁺) and S 2p peaks corresponding to the quaternary ammonium and sulfonate groups of SBMA, respectively, and the F 1s peak corresponding to TFEMA. Notably, the high-resolution XPS profiles enabled the distinction between C–N⁺ and C–N peaks observed at 401.6 and 399.7 eV, respectively (Figure 3b). The C–N⁺ to C–N ratio increased as the SBMA content in the copolymer brush increased. An increase in the S 2p peak intensity, accompanied by a decrease in the F 1s peak intensity, was also observed. These results confirm the formation of amphiphilic zwitterionic poly(SBMA_{*x*}-co-TFEMA_{*y*}) brushes with varying SBMA and TFEMA contents. Table 2 presents the compositions of the polymer brushes formed on the substrate surface as determined via XPS peak analyses. As the TFEMA monomer contains three fluorine atoms and SBMA contains one sulfur atom, the fluorine-to-nitrogen ratio (F/3N) was used to determine the contents of TFEMA and SBMA in the poly(SBMA_{*x*}-co-TFEMA_{*y*}) brushes (Table 2). The TFEMA/SBMA ratio of poly(SBMA_{*x*}-co-TFEMA_{*y*}) was consistent with the feed molar ratio of the two monomers when the SBMA/TFEMA ratios used in the synthesis were 5:5, 3:7, and 2:8. However, a large difference was observed for the brushes formed using an SBMA/TFEMA ratio of 1:9, possibly because of the different reactivities of the two monomers. The

polymer thickness decreased with an increasing TFEMA content, indicating that the polymerization rate of TFEMA was lower than that of SBMA in this system. TFE is known as a good solvent for SB polymers due to its strong hydrogen-bond donating power.^{49,50} We believe that the ability of TFE to stabilize sulfonate anions provided a better environment for growing poly(SBMA) than poly(TFEMA). Therefore, when the poly(SBMA_{*x*}-co-TFEMA_{*y*}) brushes were formed, a smaller amount of TFEMA than that expected was included in the poly(SBMA_{*x*}-co-TFEMA_{*y*}) brushes as the SBMA/TFEMA ratio was increased to 1:9.

The surface morphologies of the copolymer brushes were analyzed by using AFM (Figure 4). After pDA coating, the very smooth surface of the original substrate, with a root-mean-square (RMS) roughness of 0.1 nm, changed to a rough surface with an RMS roughness of 13.7 nm. This is due to the pDA particles formed on the surface during the pDA coating process. However, the SI-ATRP of SBMA and TFEMA did not significantly affect the surface morphology. Rough surfaces with submicrometer aggregates were observed because the formed polymer brushes were not sufficiently thick (<10 nm) to affect the surface roughness.

The marine antifouling properties of the amphiphilic zwitterionic poly(SBMA_{*x*}-co-TFEMA_{*y*}) brushes were investigated by studying the adhesion of marine diatoms to them (Figure 5). The marine diatom adhesion test was performed under dynamic conditions; in the test, the polymer brush-coated substrate was immersed in a marine diatom suspension and continuously shaken. The hydrophilic SBMA component in the polymer brush inhibited the initial adhesion of marine diatoms, while the hydrophobic TFEMA component weakened their adhesion to the solid surface. Using poly(SBMA) brushes as the control, we confirmed the excellent diatom resistance of the coated surface; poly(SBMA) caused an 84% reduction in diatom adhesion compared to that of the uncoated surface. Using this as the reference, we compared the effects of TFEMA addition on the marine fouling resistance of the amphiphilic zwitterionic polymer coating. The marine antifouling properties of the poly(SBMA) brushes were maintained in the cases of the coatings prepared using SBMA/TFEMA ratios of 5:5 and 3:7. Quantitatively, compared with that of the uncoated surface, 91 and 88% reductions in diatom adhesion, respectively, were observed for the samples coated with polymer brushes at these SBMA/TFEMA ratios. However, the marine antifouling properties were significantly reduced at SBMA/TFEMA ratios of 2:8 and 1:9, resulting in 55 and 60% reductions in diatom adhesion, respectively, compared to that of the uncoated surface. The reduction in the marine antifouling properties at SBMA/TFEMA ratios of 2:8 and 1:9 can be explained in two ways. The first reason is the significant increase in the hydrophobicity of the coating with an increase in the ratio of TFEMA in the polymer brush. When the SBMA/TFEMA ratio was changed from 3:7 to 2:8, the WCA increased from 85 to 100°. This increase in hydrophobicity is not adequate to prevent the initial adhesion of marine diatoms, and the overall diatom adhesion increased. The second reason could be the decrease in the polymer thickness. As mentioned previously, the polymer thickness decreased with an increasing TFEMA monomer content used in the synthesis. The decrease in the polymer thickness reduced the extent to which the polymer brushes covered the solid surface. Consequently, the initial adhesion of marine diatoms was not sufficiently prevented.

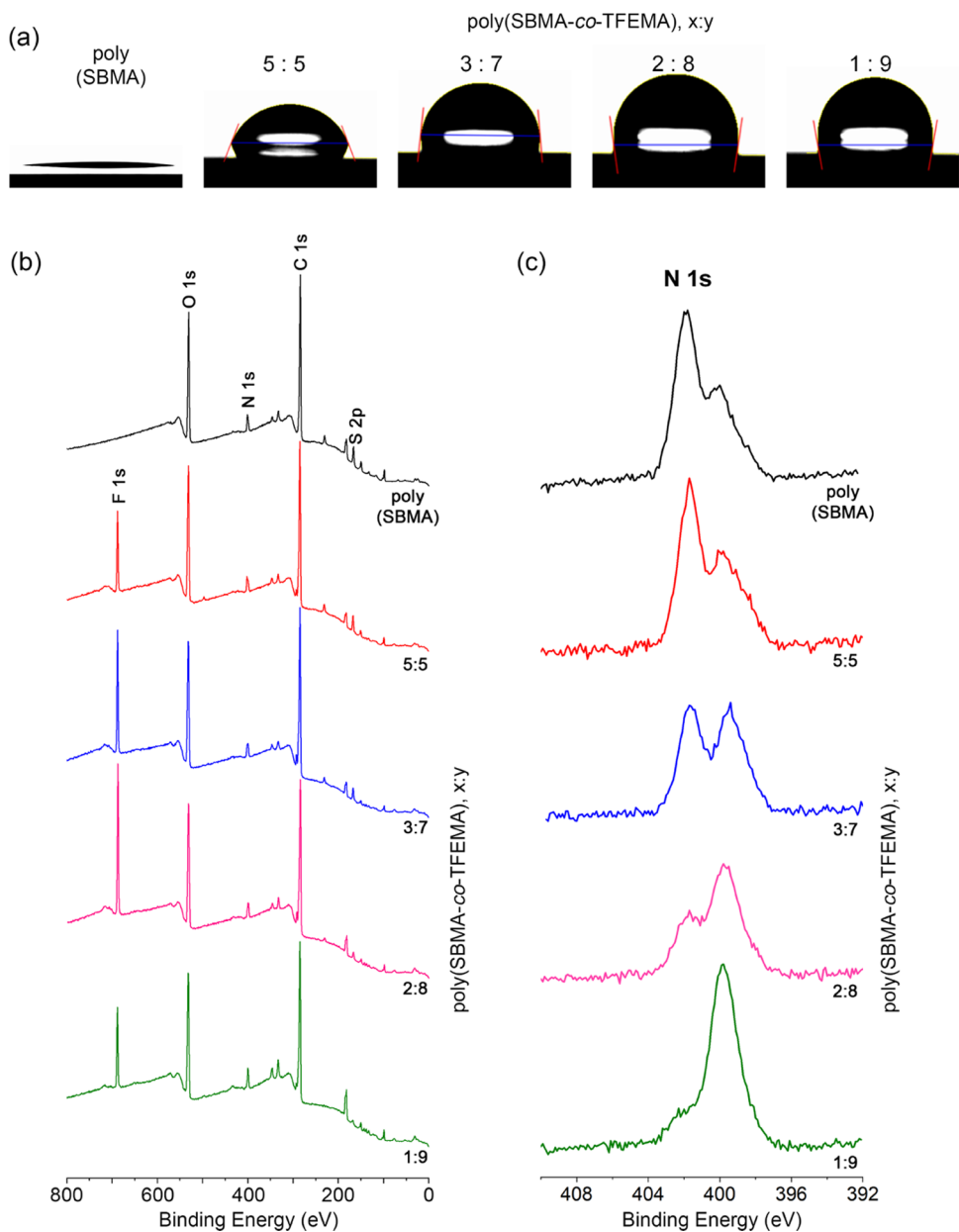


Figure 3. (a) Images showing the WCAs of Si/SiO₂ substrates coated with poly(SBMA) and poly(SBMA_{*x*}-co-TFEMA_{*y*}). (b) Survey and (c) high-resolution (N 1s) XPS profiles of the Si/SiO₂ substrates coated with poly(SBMA) and poly(SBMA_{*x*}-co-TFEMA_{*y*}).

Table 2. Thicknesses and TFEMA/SBMA Ratios of Polymer Brushes Grown on Si/SiO₂ Substrates

sample		thickness (nm)	TFEMA/SBMA ratio in the feed	TFEMA/SBMA ratio from XPS
poly(SBMA)		10.6 ± 1.6		
poly(SBMA _{<i>x</i>} -co-TFEMA _{<i>y</i>}), <i>x</i> / <i>y</i>	5:5	6.0 ± 1.0	1	0.84
	3:7	5.5 ± 1.8	2.33	1.76
	2:8	4.6 ± 0.7	4	3.22
	1:9	1.9 ± 0.8	9	3.32

Next, we investigated the resistance of poly(SBMA_{*x*}-co-TFEMA_{*y*}) brushes to sediment (silt) adsorption because a sufficient resistance to silt adsorption as well as diatom adhesion is a prerequisite for their practical application. For silt adsorption tests, poly(SBMA_{*x*}-co-TFEMA_{*y*}) brushes were formed on glass substrates. The polymer-coated samples were immersed in a silt suspension for 60 min, and the amount of silt adsorbed to the surface was determined from

the difference between the absorbances at 440 nm before and after silt adsorption (Figure 6). The highest silt adsorption was observed on the substrate coated with poly(SBMA) brushes, whereas minimal silt adsorption was observed on the substrate coated with poly(SBMA₃-co-TFEMA₇) brushes. Notably, silt adsorption increased for poly(SBMA_{*x*}-co-TFEMA_{*y*}) brushes obtained by using other SBMA/TFEMA ratios. The increase in silt adsorption at lower TFEMA contents is possibly owing to

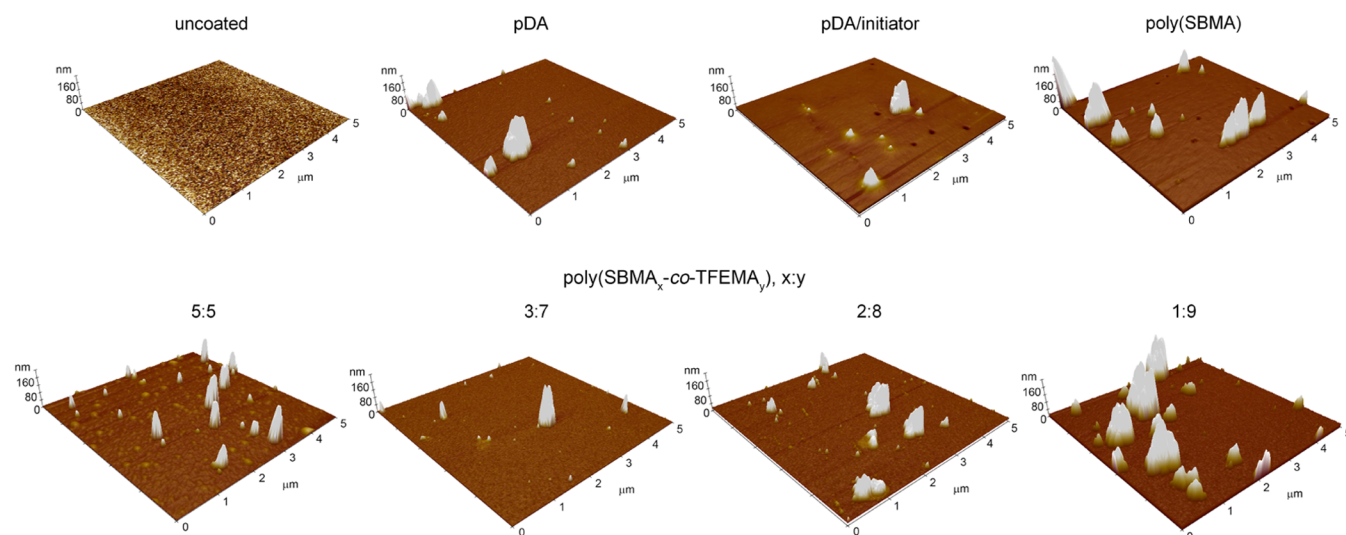


Figure 4. AFM images of the uncoated Si/SiO₂ substrate and Si/SiO₂ substrates coated with pDA, pDA/initiator, poly(SBMA), and poly(SBMA_x-co-TFEMA_y).

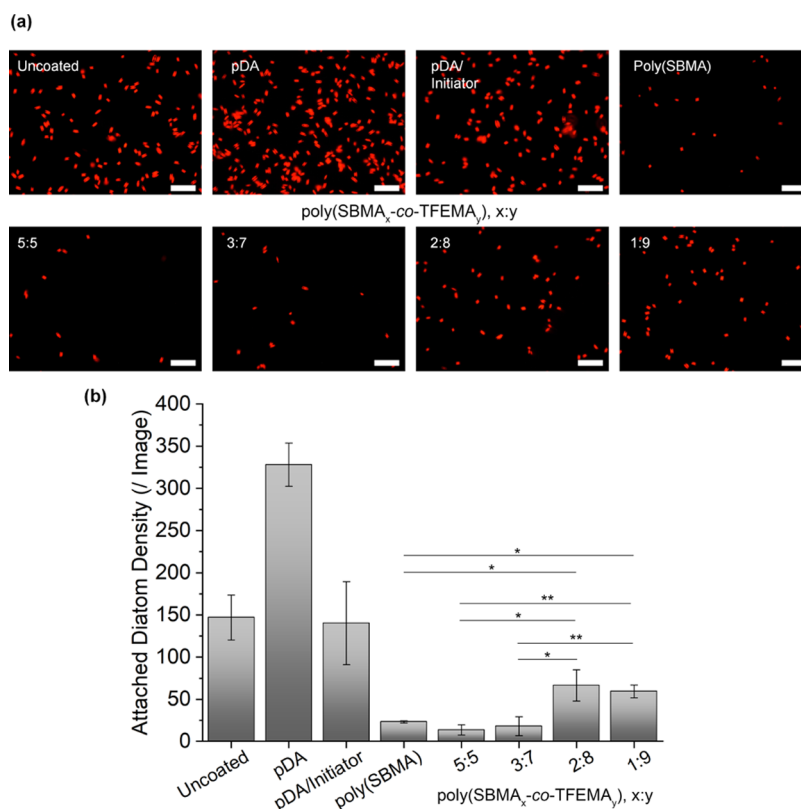


Figure 5. (a) Fluorescence images of the uncoated Si/SiO₂ substrate and Si/SiO₂ substrates coated with pDA, pDA/initiator, poly(SBMA), and poly(SBMA_x-co-TFEMA_y) and (b) quantification of the diatoms attached to them. Each value is the mean of 60 counts of three replicate samples. The error bars represent the 95% confidence limit. The scale bars represent 50 μm. Significant differences between the two groups were analyzed via unpaired Student's *t*-test. ***p* < 0.01, **p* < 0.05.

the increased proportion of the hydrophilic component (SBMA), which caused silt adsorption, whereas the increase at higher TFEMA contents was owing to a reduced polymer brush thickness.

Together, these findings demonstrate that the SI-ATRP of SBMA and TFEMA to form poly(SBMA_x-co-TFEMA_y) brushes offers an effective pathway to resolve the problems associated with the use of zwitterionic poly(SBMA) in marine

antifouling applications. Poly(SBMA₃-co-TFEMA₇) brushes coated on the substrate effectively suppress both marine diatom adhesion and silt adsorption, whereas severe silt adsorption occurs on the poly(SBMA) brush.

The long-term stability of the poly(SBMA_x-co-TFEMA_y) brushes was last investigated. For the stability test, we used the poly(SBMA₃-co-TFEMA₇) brush, which showed the best marine antifouling property, and the stability of the polymer

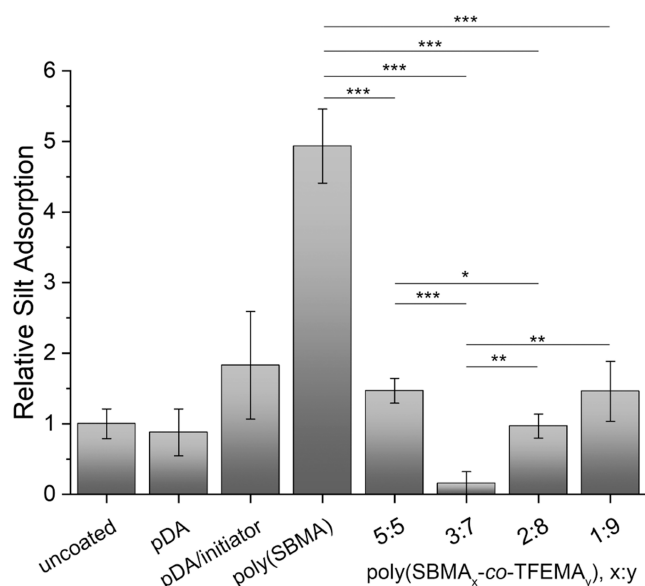


Figure 6. Silt adsorption on glass substrates with different coatings after 60 min of immersion in a sand suspension. The absorbance of the coatings at 440 nm was measured before and after immersion in the sand suspension. Each point is the 9-count mean on three replicated samples. Error bars represent the 95% confidence limit. The amount of silt adsorbed on the uncoated glass surface is considered 1. Significant differences between two groups were determined via unpaired Student's *t*-test. ****p* < 0.001, ***p* < 0.01, and **p* < 0.05.

brush was examined by measuring the thickness change upon seawater treatment. The polymer brush remained stable in seawater conditions, maintaining its thickness for up to 7 days (Figure S1). In addition to the model substrate, Si/SiO₂, the poly(SBMA₃-co-TFEMA₇) brush was also formed on SS substrates (a major element for marine vessels) because the control of marine fouling on SS is practically necessary. The poly(SBMA₃-co-TFEMA₇) brush-coated SS surface showed excellent marine antifouling performance (Figure S2). After seawater immersion for 7 days, a slight increase in the attached diatom density was observed, but the poly(SBMA₃-co-TFEMA₇) brush-coated SS surface still showed an 84% reduction of diatom adhesion compared with the uncoated SS, confirming the long-term marine antifouling performance under seawater conditions.

4. CONCLUSIONS

We developed a facile metal-ion-mediated method for immobilizing polymerization initiators on solid substrates to synthesize amphiphilic zwitterionic polymer brushes on an initiator-coated surface. Zwitterionic SBMA and hydrophobic TFEMA were used as monomers, and thin polymer brushes with different SBMA/TFEMA ratios were fabricated for marine antifouling applications. Diatom adhesion and silt adsorption tests were conducted to verify the marine antifouling performance of the polymer brushes, and the results were compared with those obtained by using the substrate coated with poly(SBMA) brushes. The poly(SBMA_x-co-TFEMA_y) brushes produced at an SBMA/TFEMA ratio of 3:7 exhibited the best marine antifouling performance and sufficiently prevented both diatom adhesion and silt adsorption. Remarkably, we found a composition that satisfied the conditions for using poly(SBMA_x-co-TFEMA_y) brushes in practical marine antifouling applications. The findings of the

study provide important insights into the design of other amphiphilic marine antifouling materials in the future.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.3c03687>.

Thickness of the poly(SBMA₃-co-TFEMA₇) brush after different incubation times in seawater; long-term stability of stainless steel substrates coated with poly(SBMA₃-co-TFEMA₇) (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This research was supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Science and ICT (NRF-2022R1A2C1005163).

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